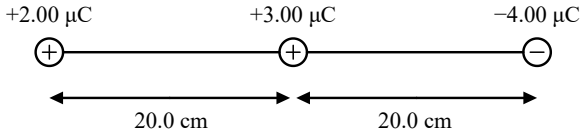
STPM 2022 PHYSICS 960/2 (U2)

Questions and Answers

Values of constants: $e = 1.60 \times 10^{-19}$ C, $\varepsilon_0 = 8.85 \times 10^{-12}$ F m^{-1}, $k = \frac{1}{4\pi\varepsilon_0} = 9.0 \times 10^{9}$ N m^2 C^{-2}, $\mu_0 = 4\pi \times 10^{-7}$ H m^{-1}. *Q19(c)(i) contains inconsistent data; both answers are shown.*

Section A [15 marks]

1. Three point charges, +2.00 μC, +3.00 μC and −4.00 μC, are placed on a straight line at a distance of 20.0 cm from one another as shown in the diagram below.



Which is the magnitude and direction of the resultant force acting on the charge +3.00 μC?

	Magnitude	*Direction*
A	1.35 N	Towards the right
B	1.35 N	Towards the left
C	4.05 N	Towards the right
D	4.05 N	Towards the left

Answer: C

From +2.00 μC (repulsion, to the right): $\frac{(9.0 \times 10^9)(2.00 \times 10^{-6})(3.00 \times 10^{-6})}{(0.200)^2} = 1.35$ N

From −4.00 μC (attraction, to the right): $\frac{(9.0 \times 10^9)(4.00 \times 10^{-6})(3.00 \times 10^{-6})}{(0.200)^2} = 2.70$ N

Both act to the right: 4.05 N towards the right.

Explanation: Find each force's direction before adding.

2. The electric field strength and the electric potential on the surface of a charged solid metal sphere are E_o and V_o respectively. Which relationship is true about the electric field strength, E, and the electric potential, V, at a point inside the sphere?

A $E = 0$ and $V = 0$

B $E = 0$ and $V = V_o$

C $E = E_o$ and $V = V_o$

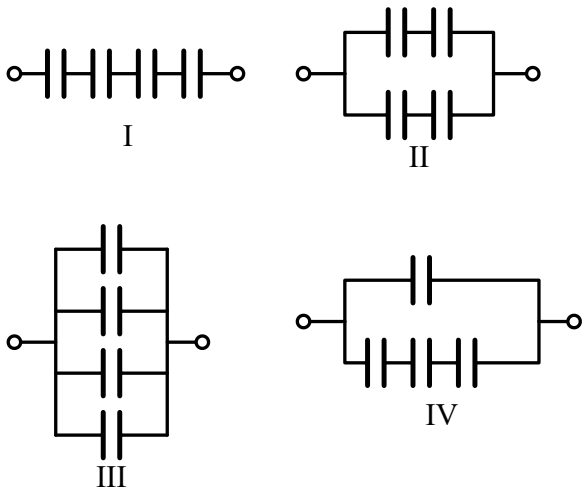
D $E = E_o$ and $V = 0$

Answer: B

Inside a conductor $E = 0$, so the potential does not change from its surface value: $V = V_o$ (B).

Explanation: $E = -\mathrm{d}V/\mathrm{d}r$: zero field means constant potential.

3. Four identical capacitors are connected in different arrangement as shown in the diagrams below.



Which sequence of arrangement shows the equivalent capacitance in descending order?

A I > IV > II > III
B I > II > IV > III
C III > I > IV > II
D III > IV > II > I

Answer: D

I: four in series = $C/4$. II: two branches of $C/2 = C$. III: four in parallel = $4C$. IV: C in parallel with three in series = $C + C/3 = 1.33C$.

Descending: III > IV > II > I.

Explanation: Parallel adds; series reduces.

4. A parallel plate capacitor is charged by a battery and then disconnected. What happens to the change in the charge, potential difference and capacitance of the capacitor when the distance between the plates is increased?

	Charge	*Potential difference*	*Capacitance*
A	Constant	Constant	Increases
B	Constant	Increases	Decreases
C	Increases	Decreases	Increases
D	Increases	Increases	Decreases

Answer: B

Disconnected: Q constant. $C = \varepsilon_0 A/d$ decreases as d increases, so $V = Q/C$ increases (B).

Explanation: The field $E = Q/\varepsilon_0 A$ stays the same; $V = Ed$ grows.

5. When a metal rod is heated, the electric conductivity of the rod

A increases because the mean time between the collisions of electrons increases
B decreases because the mean time between the collisions of electrons increases
C increases because the mean time between the collisions of electrons decreases
D decreases because the mean time between the collisions of electrons decreases

Answer: D

Heating increases lattice vibration, so the mean time between collisions decreases and the conductivity decreases (D).

Explanation: $\sigma = ne^2\tau/m$; for a metal n is almost constant.

6. A battery has an e.m.f. of 12 V and an internal resistance of 0.50 Ω. The terminals of the battery are connected to a load of resistance 5.0 Ω. What is the power dissipated by the internal resistance of the battery?

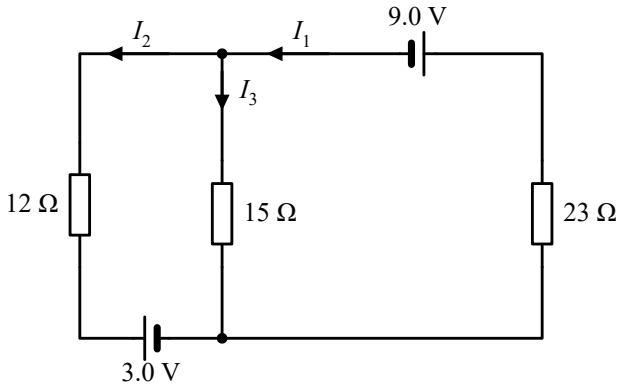
A 0.55 W B 2.4 W C 24 W D 26 W

Answer: B

$$I = \frac{12}{5.0 + 0.50} = 2.18 \text{ A}; P = I^2 r = (2.18)^2(0.50) = 2.4 \text{ W}$$

Explanation: This power is wasted as heat inside the battery.

7. A circuit consists of two d.c. supplies and three resistors are arranged as shown in the circuit diagram below.



What is the value of $\frac{I_2}{I_1}$?

A 1.2 B 0.86 C 0.76 D 0.14

Answer: B

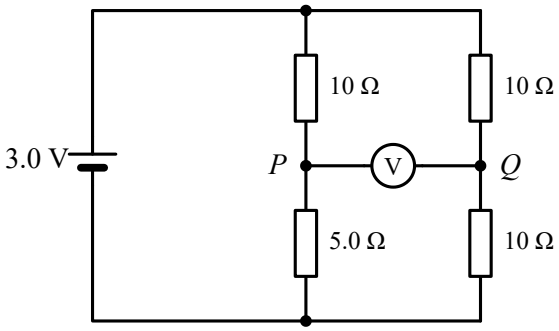
Let U = p.d. of the top junction above the bottom junction. $I_3 = U/15$, $I_1 = (9.0 - U)/23$, $I_2 = (U + 3.0)/12$ (the 3.0 V cell drives current round the left branch in the direction of I_2).

$$I_1 = I_2 + I_3 \Longrightarrow U = 0.730 \text{ V}, I_1 = 0.360 \text{ A}, I_2 = 0.311 \text{ A}$$

$$\frac{I_2}{I_1} = \frac{0.311}{0.360} = 0.86$$

Explanation: Kirchhoff's laws (loops) give the same result.

8. A battery of e.m.f. 3.0 V is connected to four resistors as shown in the circuit diagram below.



The internal resistance of the battery is negligible. What is the reading of the voltmeter?

A 0.50 V B 1.0 V C 1.5 V D 2.0 V

Answer: A

$$V_P \text{ (above the bottom wire)} = 3.0 \times \frac{5.0}{15} = 1.0 \text{ V};\ V_Q = 3.0 \times \frac{10}{20} = 1.5 \text{ V}$$

Voltmeter reading = 1.5 − 1.0 = 0.50 V.

Explanation: Treat each side as a separate potential divider (ideal voltmeter).

9. An ion of mass m and charge q moves perpendicularly to a magnetic field at a constant speed. The path of the ion in the magnetic field is

A a circle with its radius directly proportional to $\frac{m}{q}$

B a circle with its radius directly proportional to $\frac{q}{m}$

C a parabola with acceleration of the ion directly proportional to $\frac{m}{q}$

D a parabola with acceleration of the ion directly proportional to $\frac{q}{m}$

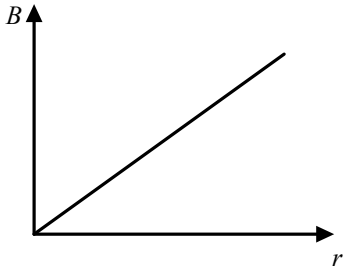
Answer: A

The force qvB is perpendicular to **v**: a circle with $r = \frac{mv}{qB} \propto \frac{m}{q}$

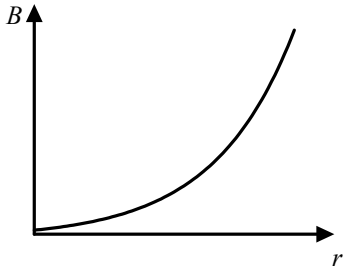
Explanation: A parabola needs a constant force, as in an electric field.

10. The strength of a magnetic flux density, B, at a perpendicular distance, r, from a straight wire which carries a current, I, is measured. Which graph shows the variation of B with r?

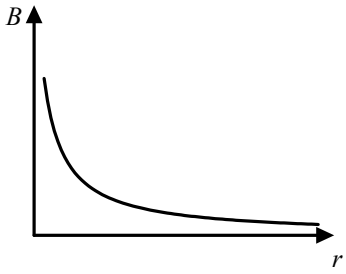
A



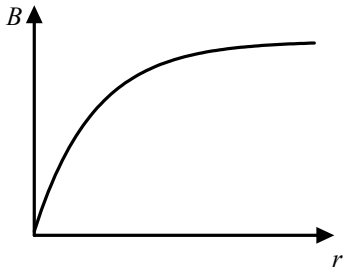
B



C



D



Answer: C

$B = \frac{\mu_0 I}{2\pi r} \propto \frac{1}{r}$: a hyperbola-shaped curve (C)

Explanation: Doubling r halves B.

11. Which is **not** true about the application of the Hall effect?

A Measures the magnetic flux density
B Identifies the type of semiconductor
C Measures the specific charge $\frac{e}{m}$ of the electrons
D Measures the density of charge carriers in semiconductor

Answer: C

The Hall effect measures B, the carrier density n and the sign of the carriers. It does not measure e/m (C).

Explanation: e/m is found with crossed electric and magnetic fields (Thomson).

12. The self inductance of a solenoid will **not** increase by

A increasing the radius of the solenoid
B decreasing the length of the solenoid
C inserting an iron rod into the core of the solenoid
D increasing the current flows in the solenoid

Answer: D

$L = \frac{\mu N^2 A}{l}$: it does not depend on the current (D)

Explanation: A larger radius, a shorter length or an iron core all increase L.

13. An inductor stores energy of 40 J when a current 5.0 A flows through it. If the current falls to zero in 0.050 s, what is the average back e.m.f. induced in the inductor?

A 100 V B 160 V C 320 V D 1600 V

Answer: C

$$\frac{1}{2}LI^2 = 40 \Longrightarrow L = \frac{2(40)}{5.0^2} = 3.2 \text{ H}$$

$$\varepsilon = L\frac{\Delta I}{\Delta t} = (3.2)\frac{5.0}{0.050} = 320 \text{ V}$$

Explanation: Find L from the stored energy first.

14. A pure inductor is connected to an alternating supply. The r.m.s. voltage of the supply is V and the r.m.s. current of the circuit is I. Which statement is true about the inductor?

A The mean power dissipation is zero and voltage leads current by $\frac{\pi}{2}$ rad.

B The mean power dissipation is zero and voltage lags current by $\frac{\pi}{2}$ rad.

C The mean power dissipation is IV and voltage leads current by $\frac{\pi}{2}$ rad.

D The mean power dissipation is IV and voltage lags current by $\frac{\pi}{2}$ rad.

Answer: A

In a pure inductor the voltage leads the current by $\pi/2$ and the mean power is zero (A).

Explanation: Energy stored in one quarter-cycle is returned in the next.

15. An alternating voltage supplied to a capacitor of capacitance 200 μF is given by $V = 5.0 \sin(300t)$, where V is in volts and t is in seconds. Which equation represents the current, I, that flows through the capacitor?

A $I = 0.21 \sin(300t)$ B $I = 0.21 \cos(300t)$

C $I = 0.30 \sin(300t)$ D $I = 0.30 \cos(300t)$

Answer: D

$$I = C\frac{dV}{dt} = (200 \times 10^{-6})(5.0)(300)\cos(300t) = 0.30\cos(300t)$$

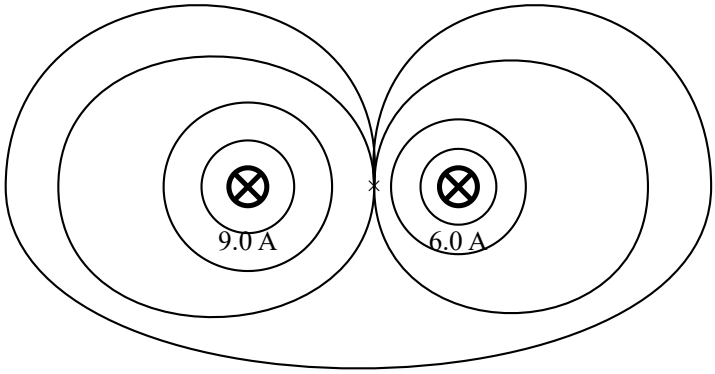
Explanation: In a capacitor the current leads the voltage by 90°.

Section B [15 marks]

16. Two parallel long wires, I_1 and I_2, carries current 9.0 A and 6.0 A respectively, in the same direction are at 10 cm apart.

(a) Sketch the magnetic field formed around the wires. [2]

Answer



neutral point between the wires, nearer the 6.0 A wire

Concentric circular field lines around each wire in the correct sense	1
Neutral point between the wires, nearer the 6.0 A wire, with outer lines enclosing both	1
	2

Explanation: Use the right-hand grip rule; with both currents into the page the lines are clockwise.

(b) Determine the magnitude of the force per unit length that acts on each wire. [2]

Answer

$\frac{F}{l} = \frac{\mu_0 I_1 I_2}{2\pi d} = \frac{(4\pi \times 10^{-7})(9.0)(6.0)}{2\pi(0.10)}$	1
$= 1.08 \times 10^{-4}$ N m^{-1} (attractive)	1
	2

Explanation: Newton's third law: the same force per unit length acts on each wire.

(c) Determine the distance of a point from I_1 if the resultant magnetic field is zero. [4]

Answer

Currents in the same direction: the fields cancel at a point between the wires	1
$\frac{\mu_0(9.0)}{2\pi x} = \frac{\mu_0(6.0)}{2\pi(0.10 - x)}$	1
$9.0(0.10 - x) = 6.0x$	1
$x = 0.060$ m $= 6.0$ cm from I_1	1
	4
Total for question	**8**

Explanation: The neutral point is nearer the smaller current.

17. A resistor of 5.00 kΩ and an inductor of 2.00 H are connected in series across an alternating supply of 60.0 Hz with peak voltage 240 V.

(a) Calculate the impedance of the circuit. [3]

Answer

$X_L = 2\pi fL = 2\pi(60.0)(2.00) = 754\ \Omega$ 1

$Z = \sqrt{(R^2 + X_L^2)} = \sqrt{((5000)^2 + (754)^2)}$ 1

$Z = 5.06 \times 10^3\ \Omega$ 1

3

Explanation: Resistance and reactance add as phasors.

(b) Calculate the peak current in the circuit. [2]

Answer

$I_0 = \frac{V_0}{Z} = \frac{240}{5057}$ 1

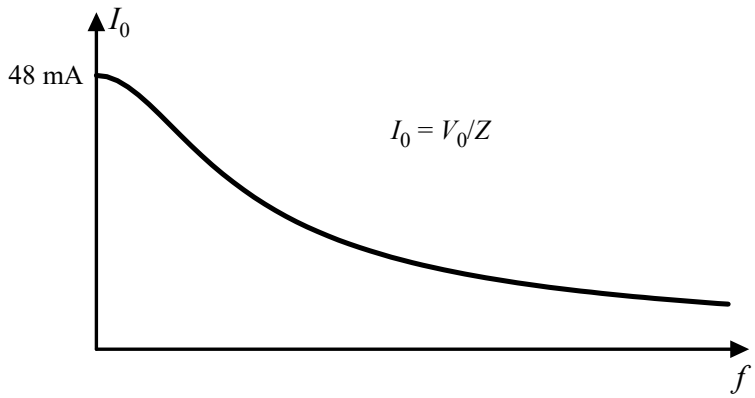
$I_0 = 4.75 \times 10^{-2}$ A (47.5 mA) 1

2

Explanation: Peak voltage gives peak current.

(c) Sketch a graph to show the variation of the peak current with the frequency of the alternating supply. [2]

Answer



Maximum value (V_0/R = 48 mA) at f = 0	1
Decreasing curve approaching zero as f increases	1
	2
Total for question	**7**

Explanation: $X_L = 2\pi fL$ grows with frequency, so Z grows.

Section C [30 marks]

18. (a) State **three** ways to increase the capacitance of an isolated parallel plate capacitor. [3]

Answer

Increase the area of overlap of the plates	1
Decrease the separation of the plates	1
Insert a dielectric (of higher dielectric constant) between the plates	1
	3

Explanation: From $C = \frac{\varepsilon_r \varepsilon_0 A}{d}$.

(b) A parallel plate capacitor, C, a resistor, R, and an e.m.f., ε, are connected in series. Derive an expression for the charge Q in the capacitor as a function of charging time, t. [6]

Answer

Kirchhoff: $\varepsilon = IR + \frac{Q}{C}$ 1

$I = \frac{dQ}{dt} \Longrightarrow R\frac{dQ}{dt} = \frac{\varepsilon C - Q}{C}$ 1

$\frac{dQ}{\varepsilon C - Q} = \frac{dt}{RC}$ 1

Integrate from $Q = 0$ at $t = 0$: $-\ln(\varepsilon C - Q) + \ln \varepsilon C = \frac{t}{RC}$ 1

$\frac{\varepsilon C - Q}{\varepsilon C} = e^{-t/RC}$ 1

$Q = \varepsilon C(1 - e^{-t/RC}) = Q_0(1 - e^{-t/RC})$ 1

6

Explanation: $Q_0 = \varepsilon C$ is the final charge.

(c) An air-filled parallel plate capacitor is separated by a distance of 3.0 mm and the plate has an area of 20 cm^2.

(i) Calculate the capacitance of the capacitor. [2]

Answer

$C = \frac{\varepsilon_0 A}{d} = \frac{(8.85 \times 10^{-12})(20 \times 10^{-4})}{3.0 \times 10^{-3}}$ 1

$C = 5.9 \times 10^{-12}$ F 1

2

Explanation: 20 $cm^2 = 20 \times 10^{-4}$ m^2.

(ii) If a mica of dielectric constant 7 is inserted between the plates, calculate the new capacitance of the capacitor. [2]

Answer

$C' = \varepsilon_r C = 7(5.9 \times 10^{-12})$ 1

$C' = 4.13 \times 10^{-11}$ F 1

2

Explanation: The mica fills the whole gap.

(iii) If the capacitor in (c)(ii) is connected to an e.m.f. of 12 V, calculate the energy stored in the capacitor. [2]

Answer

$U = \frac{1}{2}CV^2 = \frac{1}{2}(4.13 \times 10^{-11})(12)^2$ 1

$U = 2.97 \times 10^{-9}$ J 1

2

Total for question **15**

Explanation: Equivalently $Q^2/2C$ or $\frac{1}{2}QV$.

19. (a) Explain the existence of drift velocity in a conductor. [3]

Answer

Without a p.d. the free electrons move randomly; their average velocity is zero 1

A p.d. produces an electric field that accelerates the electrons opposite to the field 1

Collisions with the lattice ions repeatedly stop them, so they gain only a small average velocity in one direction: the drift velocity 1

3

Explanation: The drift velocity is superimposed on the much faster random motion.

(b) Explain the effects of an increase in temperature on the resistance of a conductor and a semiconductor. [6]

Answer

Conductor: the number of free electrons is (almost) constant 1

The ions vibrate with larger amplitude, so electrons collide more often 1

The resistance increases 1

Semiconductor: more electrons gain energy to move into the conduction band, creating electron–hole pairs 1

The number of charge carriers increases greatly, outweighing the increase in collisions 1

The resistance decreases 1

6

Explanation: This is why thermistors (NTC) are made from semiconductors.

(c) A copper wire of length 20.0 m and diameter 1.5 mm has resistivity of 1.68×10^{-8} Ω m.

(i) A current of 1.7 A flows in the wire when a potential difference of 5.0 V is applied. Calculate the drift velocity of free electrons. [4]

Answer

$A = \pi(0.75 \times 10^{-3})^2 = 1.77 \times 10^{-6}$ m^2 — 1

$v_d = \dfrac{I}{nAe}$ — 1

$v_d = \dfrac{1.7}{(8.4 \times 10^{28})(1.77 \times 10^{-6})(1.60 \times 10^{-19})}$ — 1

$v_d = 7.2 \times 10^{-5}$ m s^{-1} — 1

4

Explanation: The data are inconsistent: this wire has $R = \rho l/A = 0.190$ Ω, so 5.0 V would drive about 26 A, not 1.7 A. Using the p.d. instead, $v_d = \dfrac{V}{\rho l n e} = 1.11 \times 10^{-3}$ m s^{-1}. Either method should be accepted.

(ii) A constantan wire has a similar length and resistance with copper. If the resistivity of constantan is twice than that of copper, calculate the diameter of the constantan wire. [Given that copper wire has 8.4×10^{28} free electrons per m^3.] [2]

Answer

Same R and l: $A \propto \rho$, so A doubles and $d \propto \sqrt{(A)}$ — 1

$d = 1.5\sqrt{2} = 2.12$ mm — 1

2

Total for question **15**

Explanation: Area is proportional to the square of the diameter.

20. (a) State Faraday's law and Lenz's law. [2]

Answer

Faraday: the induced e.m.f. is directly proportional to the rate of change of magnetic flux linkage — 1

Lenz: the induced current flows in a direction that opposes the change producing it — 1

2

Explanation: $\varepsilon = -\mathrm{d}(N\Phi)/\mathrm{d}t$.

(b) Explain the phenomenon of mutual induction. [4]

Answer

Two coils are placed close together (or on a common core)	1
A changing current in the primary coil produces a changing magnetic flux	1
This changing flux links the secondary coil	1
An e.m.f. is induced in the secondary coil: $\varepsilon_2 = -M\frac{\mathrm{d}I_1}{\mathrm{d}t}$	1
	4

Explanation: This is the principle of the transformer.

(c) An air-filled solenoid of N turns has length, l, and cross-sectional area, A.

(i) Derive an expression for self-inductance, L, of the solenoid. [4]

Answer

Field inside: $B = \frac{\mu_0 NI}{l}$	1
Flux through each turn: $\Phi = BA = \frac{\mu_0 NIA}{l}$	1
Flux linkage: $N\Phi = LI$	1
$L = \frac{N\Phi}{I} = \frac{\mu_0 N^2 A}{l}$	1
	4

Explanation: L depends only on the geometry and the core.

(ii) Hence, calculate self-inductance if the solenoid has 300 turns, length 25.0 cm and cross-sectional area 4.0 cm^2. [1]

Answer

$L = \frac{(4\pi \times 10^{-7})(300)^2(4.0 \times 10^{-4})}{0.250} = 1.81 \times 10^{-4}\ \mathrm{H}$	1
	1

Explanation: 4.0 $\mathrm{cm}^2 = 4.0 \times 10^{-4}\ \mathrm{m}^2$.

(iii) If the current flow through the solenoid is 2.0 A, calculate the energy stored in the solenoid. [2]

Answer

$U = \frac{1}{2}LI^2 = \frac{1}{2}(1.81 \times 10^{-4})(2.0)^2$ 1

$U = 3.62 \times 10^{-4}$ J 1

2

Explanation: The energy is stored in the magnetic field.

(d) A metal rod moves on a parallel rail with a constant speed of 2.5 m s^{-1} perpendicular to a uniform magnetic field of 0.80 T. If the rod has a length of 1.6 m, calculate the e.m.f. produced by the rod. [2]

Answer

$\varepsilon = Blv = (0.80)(1.6)(2.5)$ 1

$\varepsilon = 3.2$ V 1

2

Total for question **15**

Explanation: The rod sweeps an area lv per second.